

DATA 505: Lab 1

Getting Started with R, RStudio and Quarto

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1 Setup & Goals

Welcome to Lab 1! In this session you will:

- Learn to navigate RStudio
- Write, save, and run an R script
- See how Quarto works (our tool for mixing text and code)

Reminder: Today is about getting comfortable with the environment.
Don't worry if the code examples feel very simple.

2 Navigating RStudio

When you open RStudio, you'll see **four panes**. The official [Getting Started guide](#) describes their purposes:

1. The Source pane is where you can edit and save R or Python scripts or author computational documents like Quarto and R Markdown.

2. The Console pane is used to write short interactive R commands.
3. The Environment pane displays temporary R objects as created during that R session.
4. The Output pane displays the plots, tables, or HTML outputs of executed code along with files saved to disk.

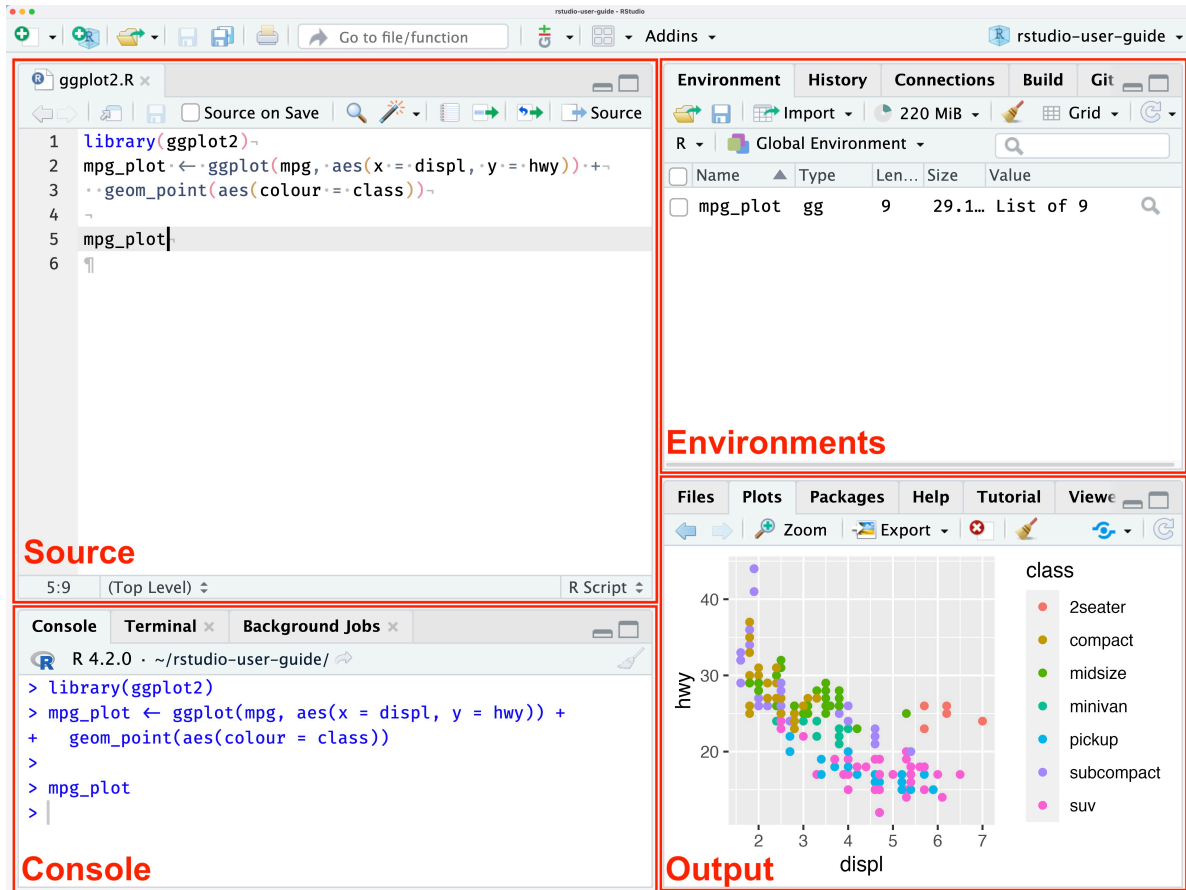


Figure 1: From [Getting Started](#) guide

2.1 Exercise

- Find the **Console** (bottom-left).
- Type:

2 + 2

3 Writing scripts and running code

The Console is for quick experiments. For work you want to save and rerun, you should use a script.

3.1 Demo

1. Create a new script:

- File -> New File -> R Script
- Using the editor window, type the code in @mean-numbers
- Save the file as `data505_lab1.R`
- You have created your first R source file!

```
c(2, 4, 6, 8, 10)
numbers <- 2 * (1:5)
print(numbers)
mean(numbers)
sample_mean <- mean(numbers)
```

2. Now you will practice executing code in RStudio. A useful reference: [“Executing Code”](#) in the RStudio User Guide

- Run the first line of @mean-numbers. What happens in the console?
- Now run the next three lines one at a time. What variables are now present in the environment?
- Next restart your R session (Session -> Restart R, or one of [these keyboard shortcuts](#)). What happened to the variables in the environment?
- Now practice running a selection of code (multiple selected lines). Select the lines and then do **Ctrl + Enter** / **Cmd + Enter** (or click “Run” from the toolbar).
- Finally, run the entire file using the “Source” button or **Ctrl + Shift + Enter**

4 Quarto demo

We will be using Quarto for lab assignments and case studies. Quarto lets us mix text (like these instructions) with code and output within a single script (a `.qmd` file).

Code is written in **code chunks**, which are executed when the document is **rendered**. The result is a formatted document which includes the code, the output, along with all the surrounding text.

We will get more practice with Quarto next week.

4.1 Exercise

- Download the Quarto script for Lab 1 from here: [lab_1_blank.qmd](#)
- Open it in RStudio
- Edit the script to add yourself as author
- Edit the code chunk below to print your name (use `print()`)
- Render the script
- Submit your edited code AND the pdf output to Moodle: